Kathuskan Thavarajah

Data Science Intern

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Profile Summary

Motivated Data Science undergraduate with a strong foundation in machine learning, statistical analysis, and natural language processing. Skilled in preparing data, developing predictive models, and communicating findings through clear visualizations. Capable of applying analytical thinking and programming skills to business problems, supporting informed decisions, and contributing to practical data-driven solutions.

Education

BSc (Hons) in Data Science **2024–present**

Sabaragamuwa University of Sri Lanka

Relevant Coursework: Machine Learning, Regression Analysis, Probability Theory, Applied Data Mining, Data Pre-Processing, Data Visualization

Technical Skills

Programming: Python, SQL, Java, JavaScript
Databases: MySQL, MongoDB
Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
Frontend Development: React.js, HTML, CSS, Tailwind
Backend Development: FastAPI, Node.js, Express.js, Laravel
Tools & Technologies: Git, GitHub, Power BI, Power Query
Soft Skills: Teamwork, Time management, Communication

Areas of Expertise

Regression Modeling, Exploratory Data Analysis, Feature Engineering, Model Evaluation, Natural Language Processing, Data Visualization, Business Intelligence

Projects

AutoValue LK - Vehicle Asking-Price Prediction | *Python, FastAPI, Scikit-learn, Pandas, HTML, CSS, JavaScript*

Individual Project - Machine Learning Web Application

- Developed a machine learning application to estimate vehicle asking prices in Sri Lanka from vehicle specifications, mileage, and condition.
- Combined and cleaned 11,750 records; performed exploratory analysis and engineered vehicle-age, brand-model, and log-transformed mileage features.
- Compared seven regression models using grouped cross-validation; selected and tuned Extra Trees, reporting held-out $R^2 = 0.739$ and 84.95% of predictions within 20% of advertised prices.
- Connected the selected preprocessing and model pipeline to a FastAPI backend, adding input validation, guided vehicle selection, and price estimates in rupees and lakhs.

LitPredict - NLP Book Rating Predictor | *Python, Scikit-learn, Pandas, TF-IDF, VADER, Random Forest*

Individual Project - NLP Regression Pipeline

- Built an NLP regression pipeline to predict average book ratings from marketing descriptions using metadata for more than 4,700 books.
- Normalized raw descriptions and combined sparse TF-IDF features with dense VADER sentiment scores for model training.
- Performed exploratory analysis and computed Pearson correlation matrices to examine relationships between text-derived features, sentiment, and average ratings.
- Trained and serialized a Random Forest regressor, achieving an MAE of 0.17 rating points, and developed a reusable training and prediction workflow.

Sri Lanka Automotive Market Analysis Dashboard | *Power BI, Power Query, Data Visualization*

Individual Project - Three-Page Interactive Dashboard

- Created a three-page Power BI dashboard to analyze Sri Lankan vehicle listings sourced from ikman.lk and Riyasewana through Kaggle datasets.
- Visualized brand composition and fuel-type distribution; identified Toyota and Suzuki as accounting for more than 50% of listings in the analyzed dataset.
- Explored asking-price relationships with manufacturing year, mileage, and location, using scatter plots to inspect pricing patterns and potential outliers.
- Built an inventory explorer with interactive range filters, supporting detailed comparison of vehicle listings across the dashboard.

IBM Stock Price Forecasting with Databricks | *Python, Databricks, PySpark, Pandas, Statsmodels*

Individual Project - Time-Series Analysis and Forecast Evaluation

- Analyzed 16,272 historical IBM stock-price records using Databricks and PySpark for data inspection, followed by statistical analysis in Python.
- Investigated price trends, returns, and rolling volatility using visualizations, stationarity tests, and autocorrelation diagnostics.
- Compared three ARIMA models against a previous-close baseline using chronological training and validation periods; selected the baseline based on its lower validation MAE.
- Evaluated the selected forecast on 414 subsequent trading days and assessed prediction intervals, identifying 83.33% coverage against a nominal 90% target.

Certificates

AI Agents 101: Building AI Agents with MCP and Open-Source Inference - AMD AI Academy **[Link]**

Knowledge Check: Get Started with Databricks for Data Engineering - Databricks Academy **[Link]**

Foundations of Business Intelligence - Coursera **[Link]**

Foundations: Data, Data, Everywhere - Coursera **[Link]**

Diabetes Disease Detection with XG-Boost and Neural Networks - Coursera **[Link]**

Leadership Journey

Prefect | Paddiruppu National School, Kaluwanchikudy

- Served as a Prefect, assisting in maintaining discipline, supporting teachers, and guiding junior students, fostering responsibility and leadership skills.

Media Team Coordinator | Uthayatharagai Arts and Sports Club, Kovil Porativu

- Coordinate media activities for the club, including content creation, event coverage, and social media management, enhancing communication and organizational skills.

Volunteering and Extracurricular

Graphic Design Volunteer | Faculty of Computing, IEEE Computer Society Chapter & Sabra Cricket Team

- Created engaging visual content, social media graphics, and promotional materials supporting academic workshops, chapter events, and university cricket activities.

Active Member | Zero Plastic Movement, Leo Club, Rotaract Club & AWS Student Builder Group (SUSL)

- Contributed to community service, environmental awareness campaigns, and social projects while developing leadership skills across multiple university organizations.

References

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